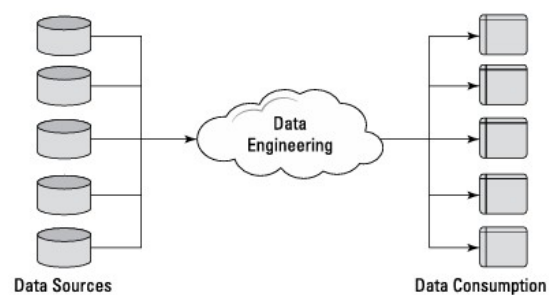


BIG DATA AND DATA ENGINEERING

- Data Era
- Data revolution
- Transform data in Opportunities
- Data-Driven Approaches

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DATA ENGINEERING



[Data engineering](#) has become one of the most critical roles in tech today. The work data engineers do enables companies to capture, process, and analyze massive amounts of data, ensuring it's high quality, accurate, and readily available and with so much change in the area, some core principles remain constant, forming the core of this role.

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- **Data Engineering/Data Science**
- **Data Pipeline/ETL pipeline**
- **Data architecture**
- **Data latency**
- **DataOps**

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DATA ENGINEERING

- **Data engineering** encompasses a broad set of procedures, tools, and skill sets that govern and facilitate the flow of data. Data engineers are experts at making data ready for consumption by working with multiple systems and tools.
- **Data architecture** is a critical aspect of successful data engineering. It lays the foundation for how data is collected, stored, processed, and utilized across an organization. With a well-designed data architecture, businesses can achieve seamless data integration, high performance, scalability, and effective data governance.
- **Data latency** is the time delay between when data generation ed and when it is available for use.
- **DataOps** (data operations) from agile programming methods to ensure the delivery of data via a continuous plan, develop, build, manage, test, release, deploy, operate, and monitor loop. Include tools, processes, and frameworks that recognize the interconnected nature of both business and IT personnel in these data-driven endeavors.

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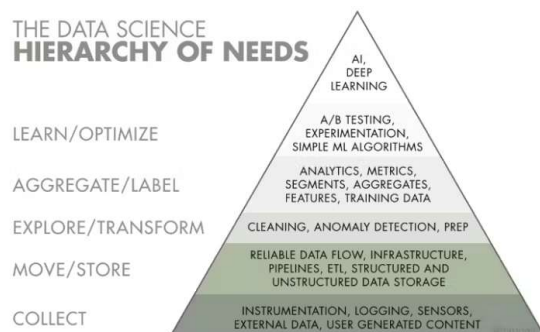
DATA ENGINEERING/DATA SCIENCE/DATA ANALYTICS

- **Data scientists** use machine learning, data exploration and other academic fields to predict future outcomes. Data science is an interdisciplinary field focused on making accurate predictions through algorithms and statistical models. Like data engineering, data science is a code-heavy role requiring an extensive programming background.
- **Data analysts** examine large datasets to identify trends and extract insights to help organizations make data-driven decisions today. While data scientists apply advanced computational techniques to manipulate data, data analysts work with predefined datasets to uncover critical information and draw meaningful conclusions.
- **Data engineers** are software engineers who build and maintain an enterprise's data infrastructure—automating data integration, creating efficient data storage models and enhancing data quality via pipeline observability. Data scientists and analysts rely on data engineers to provide them with the reliable, high-quality data they need for their work.

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BIG DATA AND DATA ENGINEERING

THE DATA SCIENCE HIERARCHY OF NEEDS



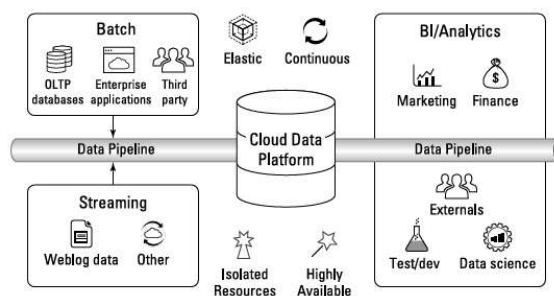
(Source: Gigster, 2023)

- **Data engineering is the discipline that creates data collection, storage, transformation, and analysis processes for large amounts of raw data, structured data, semi-structured data, and unstructured data - Big Data so that data science professionals can draw valuable insights from it.**
- **Data engineers** are the foundation that enable data scientists to perform their analysis.
- Data engineering also encompasses:
 - data quality and data access assurance.
 - design and build systems for gathering and storing data at scale as well as preparing it for further analysis.

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DATA PIPELINE

- **Scalable performance.** The ability to ingest and transform data without impacting performance or experiencing resource contention.
- **Extensible but based on standards.** Data engineers should be able to choose from a variety of languages and tools.
- **Batch and streaming data.** A complete data pipeline should support a range of data ingestion and integration options, including batch data loading and replication, as well as streaming ingestion and integration.



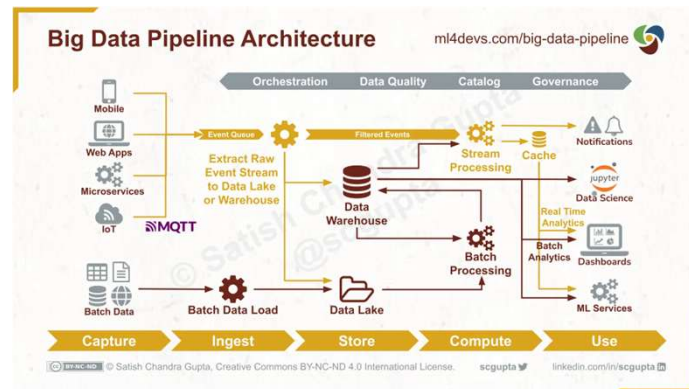
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TYPES OF DATA PIPELINES

- Batch processing pipelines
- Streaming data pipelines
- Data integration pipelines
- Cloud-native pipelines

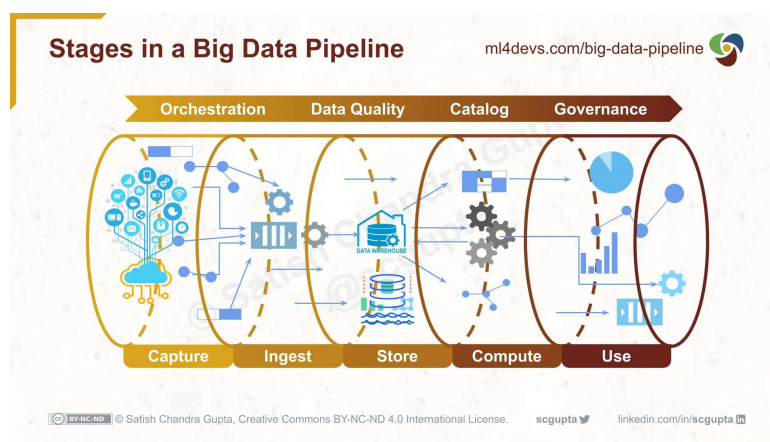
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BIG DATA PIPELINE



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BIG DATA AND IOT



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BIG DATA ARCHITECTURE CHALLENGES

- **Security:** When it comes to static big data, the data lake is the norm. Because security is required to safeguard your data from intrusion and theft, robust security is required. In addition, setting up secure access can be difficult.
- **Complexity:** The moving parts of a Big Data architecture typically consist of many interlocking elements. These components may have their own data-injection pipelines and various configuration settings to improve performance, in addition to many cross-component configuration interventions. Big Data procedures are demanding in nature and require a lot of knowledge and skill.
- **Evolving technologies:** Choosing the right solutions and components is critical to meeting Big Data business objectives. It can be challenging to determine which Big Data technologies, practices,
- **Expertise in a specific domain:** As Big Data APIs built on mainstream languages gradually become popular, we can see Big Data architectures and solutions using atypical, highly specialized languages and frameworks.

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DATA ENGINEERING PIPELINE

A **data pipeline** combines tools and operations that move data from one system to another for storage and further handling. Constructing and maintaining data pipelines is the core responsibility of data engineers. Data engineers write scripts to automate repetitive tasks – *jobs*.

- **data migration** between systems or environments (from on-premises to cloud databases);
- **data wrangling** or converting raw data into a usable format for analytics, BI, and machine learning projects;
- **data integration** from various systems and IoT devices; and
- copying tables from one database to another.

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DATA ENGINEERING PROCESS



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- **Data ingestion (acquisition)** moves data from multiple sources — SQL and NoSQL databases, IoT devices, websites, streaming services, etc. — to a target system to be transformed for further analysis. Data comes in various forms and can be both structured and unstructured.
- **Data transformation** adjusts disparate data to the needs of end users. It involves removing errors and duplicates from data, normalizing it, and converting it into the needed format.
- **Data serving** delivers transformed data to end users — a BI platform, dashboard, or data science team.
- **Data flow orchestration** provides visibility into the data engineering process, ensuring that all tasks are successfully completed. It coordinates and continuously tracks data workflows to detect and fix data quality and performance issues.

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DATA ENGINEERING KEY ELEMENTS

- **Data extraction/collection:** As its name implies, this element involves the creation of systems and processes to extract data of varying formats from multiple sources. This includes everything from structured customer data in relational databases and data warehouses, to semi-structured data such as email and website content stored on a server, and unstructured data including video, audio, and text files stored in a data lake. The variety of data formats and data sources are literally endless.
- **Data ingestion:** Data ingestion involves data source identification as well as data validation, indexing, cataloging, and formatting. Given the massive amounts of data involved, data engineering tools and data processing systems are often used to speed the ingestion of these large datasets.
- **Data storage:** Data engineers take ingested data and design the necessary data storage solutions to house it. These solutions include everything from a cloud data warehouse to a data lake, or even a NoSQL (not only structured query language) database. In addition, data engineers can also be responsible for data management within these storage solutions depending on organizational staffing and structure.
- **Data transformation:** To make data useful for data scientists as they build machine learning algorithms, as well as for use in business intelligence and data analytics, the data needs to be cleaned, enriched, and integrated with other sources. For this reason, data engineers develop ETL (extract, transform, load) data pipelines and data integration workflows to prepare these large datasets for data analysis and modeling. A variety of data engineering tools are utilized (e.g., Apache Airflow, Hadoop, Talend) depending upon the data engineer's data processing needs and the end user's (e.g., data analysts, data scientists) requirements. The final step in data transformation is to load the processed data into systems that enable data scientists, data analysts, and business intelligence professionals to work with it to produce valuable insights.
- **Data modeling, scaling, and performance:** Creating and defining data models is another key element of data engineering. In the past few years, artificial intelligence (AI) via machine learning models has become a common way to optimize everything from data volume and query load management to overall database performance and scaling infrastructure.
- **Data quality and governance:** Making sure that data is accurate and accessible is another key element of data engineering. Data engineers create validation rules and processes to ensure that organizational data governance policies are adhered to and data integrity is maintained.
- **Security and compliance:** Data engineers are often responsible for ensuring security measures prescribed by organizational cybersecurity protocols and/or industrial data privacy regulations (e.g., HIPAA) are met and all systems are in compliance.

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BIG DATA TOOLS DATA PIPELINES: ETL/ELT

When building a pipeline, a data engineer automates the data integration process with scripts—lines of code that perform repetitive tasks. Depending on their organization's needs, data engineers construct pipelines in one of the formats:

- **ETL: extract, transform, load** - ETL pipelines automate the retrieval and storage of data in a database. The raw data is extracted from the source, transformed into a standardized format by scripts and then loaded into a storage destination. ETL is the most commonly used data integration method, especially when combining data from multiple sources into a unified format.
- **ELT: extract, load, transform** - ELT pipelines extract raw data and import it into a centralized repository before standardizing it through transformation. The collected data can later be formatted as needed on a per use basis, offering a higher degree of flexibility than ETL pipelines.

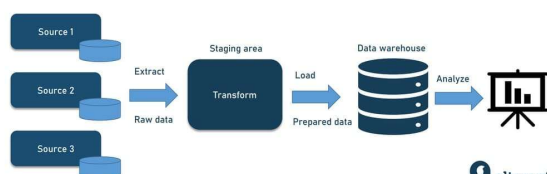
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ETL PIPELINE

ETL (Extract, Transform, Load) pipeline is the most common architecture:

1. **Extract** — retrieving data. At the start of the pipeline, we're dealing with raw data from numerous sources — databases, APIs, files, etc.
2. **Transform** — standardizing data. Having data extracted, scripts transform it to meet the format requirements. Data transformation significantly improves data discoverability and usability.
3. **Load** — saving data to a new destination. After bringing data into a usable state, engineers can load it to the destination, typically a database management system (DBMS) or data warehouse.

ETL PIPELINE

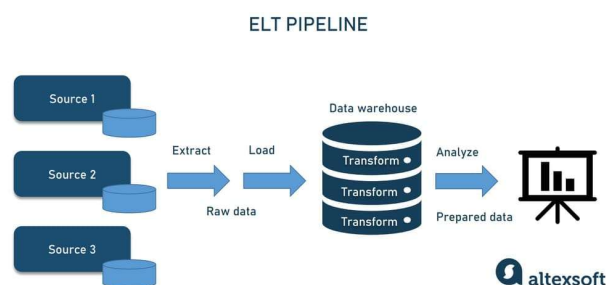


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ELT PIPELINE

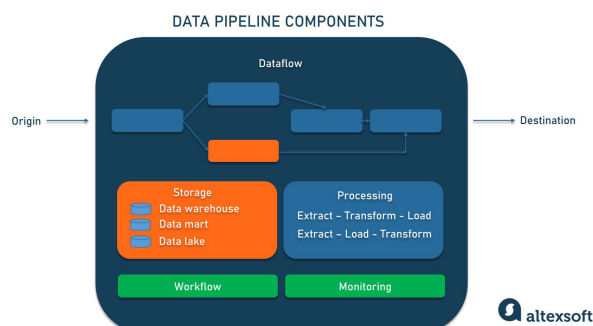
- An ELT pipeline performs the same steps but in a different order — Extract, Load, Transform. Instead of transforming all the collected data, you place it into a data warehouse, data lake, or [data lakehouse](#). Later, you can process and format it fully or partially, once or numerous times.
- ELT pipelines are preferable when you want to ingest as much data as possible and transform it later, depending on the needs arising. Unlike ETL, the ELT architecture doesn't require you to decide on data types and formats in advance. In large-scale projects, two types of data pipelines are often combined to enable both traditional and [real-time analytics](#). Also, two architectures can be involved to support [Big Data analytics](#).



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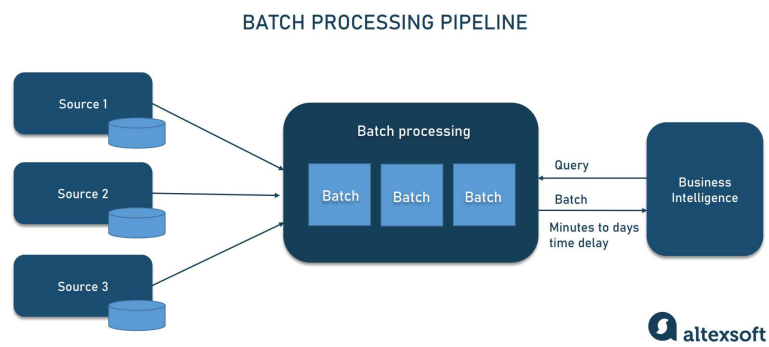
DATA PIPELINE CHALLENGES

- Setting up secure and reliable data flow is challenging. Many things can go wrong during data transportation: Data can be corrupted, hit bottlenecks causing latency, or data sources may conflict, generating duplicate or incorrect data.
- Getting data into one place requires careful planning and testing to filter out junk data, eliminating duplicates and incompatible data types to obfuscate sensitive information while not missing critical data.
 - lacking relevant metrics and
 - underestimating data load.



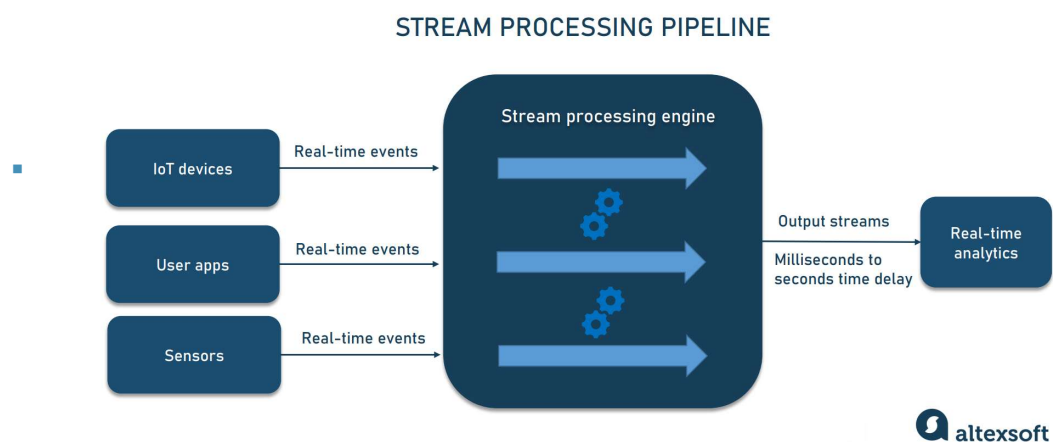
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BATCH PIPELINE FOR TRADITIONAL ANALYTICS



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STREAMING DATA PIPELINE FOR REAL-TIME ANALYTICS



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PIPELINE FOR BIG DATA ANALYTICS

- Big Data pipelines perform the same tasks as their smaller counterparts. What differentiates them is the ability to support Big Data analytics which means:
 - handling huge volumes of data,
 - coming from multiple (100+) sources
 - in a great variety of formats (structured and unstructured and semi-structured), and
 - at high speed.
- ELT loading infinite amounts of raw data and streaming analytics, extracting insights on the fly, seem to be perfect for a Big Data pipeline. Yet, thanks to modern tools, batch processing and ETL can also cope with massive amounts of information. Typically, to analyze Big Data, organizations run both batch and real-time pipelines, leveraging a combination of ETL and ELT along with several stores for different formats.

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TOPICS

- Data Lake
- Data Warehouse
- Data Lakehouse
- Data Mesh
- Data Mart
- Data Fabric

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DATA ENGINEERING: DICTIONARY

- **Data Lake** - central repository that stores raw, unstructured
- **Data Warehouse** - a structured storage system optimized for querying and reporting, typically for business intelligence (BI) purposes.
- **Data Lakehouse** - a hybrid approach that combines the benefits of a data lake and a data warehouse, allowing both raw and structured data in one platform.
- **Data Mesh** - a decentralized approach to data architecture where data is treated as a product, managed by domain-specific teams.
- **Data Fabric** - s an architecture designed to facilitate the end-to-end integration of various data pipelines and cloud environments through intelligent and automated systems. This architecture addresses the challenges posed by the exponential growth of big data, such as data silos, security risks, and decision-making bottlenecks.

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Data Lake vs Data Warehouse vs Data Lakehouse vs Data Mesh

	Description	Advantages	Limitations
Data Lake	A central repository that stores raw, unstructured, and structured data at any scale.	Highly scalable, cost-effective for storing vast amounts of data, flexible for data scientists and engineers.	Data governance can be challenging, and querying large datasets can be inefficient without proper management.
Data Warehouse	A structured storage system optimized for querying and reporting, typically for business intelligence (BI) purposes.	storage system optimized for querying and reporting, typically for business intelligence (BI) purposes. - Pros: Optimized for performance, ideal for complex queries, and provides strong data governance.	Less flexible for unstructured data, higher costs, and slower adaptation to real-time data.
Data Lakehouse	A hybrid approach that combines the benefits of a data lake and a data warehouse, allowing both raw and structured data in one platform.	Flexibility of data lakes with the performance of data warehouses, simplified architecture, and supports advanced analytics	Emerging technology with fewer mature solutions compared to standalone systems.
Data Mesh	A decentralized approach to data architecture where data is treated as a product, managed by domain-specific teams.	Promotes scalability and ownership, supports cross-functional collaboration, and enhances agility.	Requires a cultural shift, complexity in governance, and a steep learning curve.

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BIG DATA: DATA STORAGE SOLUTIONS

The systems that data engineers create begin and end with data storage solutions: harvesting data from one location, processing it and then depositing it elsewhere at the end of the pipeline.

- **Cloud computing services** - Microsoft Azure Data Lake Storage, Amazon S3 and other AWS solutions, Google Cloud and [IBM Cloud®](#) are all popular platforms.
- **Relational databases** - MySQL and PostgreSQL as two of the leading open source RDBMS options.
- **NoSQL databases**
- **Data warehouses**
- **Data lakes**
- **Data lakehouses**

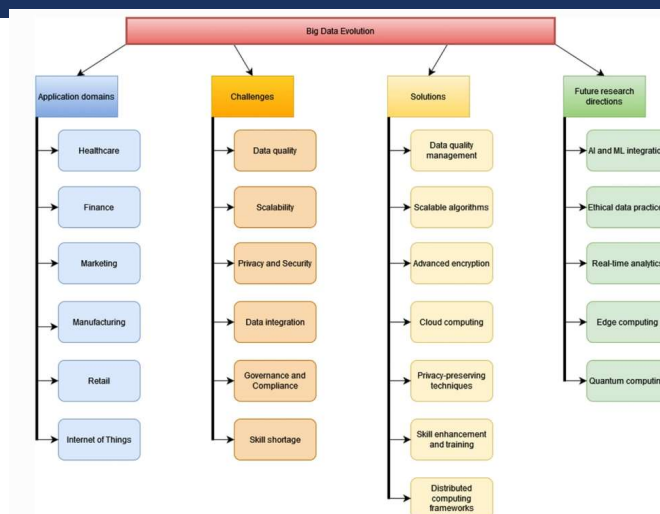
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BIG DATA: PROGRAMMING LANGUAGES

- **SQL** or structured querying language, is the predominant database creation and manipulation programming language. It forms the basis for all relational databases and may be used in NoSQL databases as well.
- **Python** offers a wide range of prebuilt modules to speed up many aspects of the data engineering process, from building complex pipelines with Luigi to managing workflows with Apache Airflow. Many user-facing software applications use Python as their foundation.
- **Scala** is a good choice for use with big data as it meshes well with Apache Spark. Unlike Python, Scala permits developers to program multiple concurrency primitives and simultaneously execute several tasks. This parallel processing ability makes Scala a popular choice for pipeline construction.
- **Java** is a popular choice for the backend of many data engineering pipelines. When organizations opt to build their own in-house data processing solutions, Java is often the programming language of choice. It also underpins Apache Hive, an analytics-focused warehouse tool.

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TAXONOMY OF BIG DATA EVOLUTION OVER THE LAST 15 YEARS



Taxonomy of Big Data evolution over the last 15 years

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